2 Symbolic Logic

This draws from Chapter 2 of the text.

Recall that an integer is prime if it is greater than one, and its only positive factors are one and itself. The following is a (sketch of a) well known proof of the fact that there are infinitely many primes.

Proof. Assume that there are only finitely many primes. Then we may list them: $p_1, p_2, \ldots, p_n$. Let $P = (p_1 \cdot p_2 \cdot \cdots \cdot p_n) + 1$. P is greater than all of the p_i so it is not prime. Since it is not prime, it can be written as the product of primes, (this fact is called the fundamental theorem of arithmetic). In particular, it can be written as $P = Qp_i$ for some p_i. But then p_i divides P and $P - 1$ so divides $P - (P - 1) = 1$. This is impossible, because $p_i \geq 2$. Our assumption that there are only finitely many prime must, therefore, be false. So there are infinitely many primes. $\square$

We wanted to prove a statement 'S is true.' but we instead assumed 'S is false' and showed this was inconsistent. This is called a 'proof by contradiction', and is a common proof technique. Similarily to prove an implication 'if a then b', a useful approach is to show that 'if b is not true, then a is not true'. Why is this okay?

Symbolic logic is useful in deciding if two approaches to a proof are equivalent. We look at symbolic logic with a view to proof, techniques, but take it further than this.

Topics Covered:

- Truth value of primitive and compound statement through truth tables
- Tautology, contradiction, logical equivalence, laws of logic, rules of substitution
- Contrapositive, inverse, converse
- Valid argument, premise, conclusion, logical implication, rules of inference
- Quantification, open statement, universe of discourse

2.1 Basic Connectives and Truth Tables

Definition 2.1. A *statement* or *proposition* is a declarative sentence that is either true or false, not both, and it must be decidable which it is.

Examples of statements are '$1 + 1 = 3$', '7 is a quadratic residue modulo 10'. The sentence '$x > 0$' is not a statement, it is neither true or false, but if x is defined as a number, then it becomes a statement.

Statements are usually denoted by lowercase letters. We might let p be the statement $1+1 = 3$, in which case we could determine that p is a false statement.

Definition 2.2. Given a statement p, the *negation* of p, or 'not p', is denoted $\neg p$.

Where p is $1+1 = 3$, $\neq p$ is the statement '$1+1 \neq 3$', which is a true statement.

2.1.1 Basic Connectives

Negation is not the only way we can construct one statement from others. A statement that cannot be broken down into smaller statements is called a *primitive statement* or just a *primitive*.

Given primitives p and q we can construct *compound statements* using any of the following logical connectives. 'And' denoted $\wedge$; 'or' denoted $\vee$; 'xor' or 'exclusive or' denoted $\underline{\vee}$; 'implies' denoted $\rightarrow$; and 'biconditional' or 'if and only if' or 'iff' denoted $\leftrightarrow$.

Statements can be build up from other statements. For example, where p is $1 + 1 = 3$ and q is $1 + 1 = 2$, $p \vee q$ is the statement that '$1 + 1$ is 3 or 2', and is true because at least one of p and q was true.

The truth value, that is, the truth or falsity, of a compound statement depends only on the truth value of the primitives in it. The above connectives are thus rigourously defined by the following table, where we denote truth by 1 and falsity by 0.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \underline{\vee} q$	$p \rightarrow q$	$p \iff q$
0	0	1	0	0	0	1	1
0	1	1	0	1	1	1	0
1	0	0	0	1	1	0	0
1	1	0	1	1	0	1	1

This table is called a *truth table*. They can be used to determine the value of any compound statement from the value of their component primitives. For more complicated statements, we build up the truth table in several steps. The general approach is to approach a statement beginning with 'not' and then bracketed statements, and then working left to right.

Problem 2.1. Make a truth table for the statement $(\neg p \vee q) \to r$.

Solution

p	q	r	$\neg p$	$\neg p \vee q$	$\neg p \vee q \to r$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	1	1

Observe that in the above example, the statement was true in seven out of eight of the possible truth assignments of the primitives. One thing we are interested in is when a statement is always true, or always false, whatever the truth values of the primitives. From now on, we treat primitives as variable, which can be true or false. Though they are true or false, we don't know what they are.

Definition 2.3. A *tautalogy* is a statement that is always true. A *contradiction* is a statement that is always false.

Example 2.4. The statement $p \vee \neg p$ is a tautology and the statement $p \wedge \neg p$ is a contradiction.

Problem 2.2. Using a truth table, decide whether or not the statement $(p \to (q \to p))$ is a tautology or a contradicion. Can you argue this without a truth table?

One sees that if a statement has n primitives, then the truth table will have 2^n rows. Three primitives is about the most we want to deal with. From here the theory of logical systems that we start to develop will allow us to decide if a statement is a truth or contradiction through more efficient means.

Remark 2.5. Denoting truth by 1 and falsity by 0 exemplifies the analogy between the logic algebra that we are developing, and a boolean algebra. We could further view $\underline{\vee}$ and $\wedge$ as analogous to addition and multiplication. We won't follow this analogue any more, but if you are familiar with boolean algebras, it might be a useful analogue.

2.2 Logical Equivalence

Problem 2.3. Construct truth tables for the statements '$p \to q$' and '$\neg p \vee q$'.

For every truth assignment of the primitives p and q, these two compound statements have the same truth value.

Definition 2.6. Statements s_1 and s_2 are *logically equivalent* if s_1 is true if and only if s_2 is true. We denote this $s_1 \iff s_2$. If p is a tautalogy, we write $p \iff T_0$ and if it is a contradiction, we write $p \iff F_0$.

Remark 2.7. The notations '$s_1 \leftrightarrow s_2$' and '$s_1 \iff s_2$' mean different things. The first is a logical statement- it may be true or false, and writing it down we are making no assertion about its truth value. The second, '$s_1 \iff s_2$', says that the first, '$s_1 \leftrightarrow s_2$', is a tautalogy.

It is easy to show (though not always easy to determine) that non-equivalent statements are non-equivalent. One only has to provide one line of the truth table- a true assignment of the primitives in which the statements differ.

Example 2.8. Show that $p \wedge (\neg p \to q)$ is not a tautalogy.

Solution

Where $p = 1$ and $q = 0$ we have $\neg p = 1$ so $(\neg p \to q) = 0$ so $p \wedge (\neg p \to q) = 1 \wedge 0 = 0$. Thus the statement is not a tautalogy.

To prove two statements are equivalent, we can always use a truth table, but we would like to avoid doing this too often. So we build up a list of equivalences that we can use over and over.

Above, we saw that

$$p \to q \iff \neg p \vee q$$

so '$\to$' can be expressed in terms of '$\neg$' and '$\vee$'. In the same way, we have

$$\begin{aligned} p \leftrightarrow q &\iff (p \to q) \wedge (q \to p) \\ &\iff (\neg p \vee q) \wedge (\neg q \vee p) \\ &\iff (\neg p \wedge \neg q) \vee (p \wedge q) \end{aligned}$$

and

$$p \underline{\vee} q \iff (p \vee q) \wedge \neg(p \wedge q).$$

Often we will use these equivalences to reduce compound statements to statements using only $\neg$, $\vee$ and $\wedge$.

Problem 2.4. Convince yourself of the logical equivalences given above. Prove that $\wedge$ can also be expressed in terms of '$\neg$' and '$\vee$'.

There are many other useful logical equivalences

Problem 2.5. Use a truth table to show that

$$\neg(p \wedge q) \iff (\neg p \vee \neg q).$$

This is called DeMorgan's law.

We will frequently use the following Laws of Logic.

1	$\neg\neg p \iff p$	Double Negation
2	$\neg(p \vee q) \iff \neg p \wedge \neg q$ $\neg(p \wedge q) \iff \neg p \vee \neg q$	Demorgan
3	$p \vee q \iff q \vee p$ $p \wedge q \iff q \wedge p$	Commutativity
4	$p \vee (q \vee r) \iff (p \vee q) \vee r$ $p \wedge (q \wedge r) \iff (p \wedge q) \wedge r$	Associativity
5	$p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$	Distributivity
6	$p \vee p \iff p$ $p \wedge p \iff p$	Idempotence Idempotence
7	$p \vee F_0 \iff p$ $p \wedge T_0 \iff p$	Identity
8	$p \vee \neg p \iff T_0$ $p \wedge \neg p \iff F_0$	Inverse
9	$p \vee T_0 \iff T_0$ $p \wedge F_0 \iff F_0$	Domination
10	$p \vee (p \wedge q) \iff p$ $p \wedge (p \vee q) \iff p$	Absorption
(11)	$p \to q \iff \neg p \vee q$	Implication

Problem 2.6. Read through each of the laws of logic, and convince yourself that they are true. Prove your favourite one using a truth table. Prove that it is your favourite one. Prove that it doesn't pay to choose favourites.

2.2.1 Substitution Rules

On top of the laws of logic, we have some substitution rules which allow us to use them more easily.

- If a statement s is a tautalogy, and every occurence of some primitive p in s is replaced by another statement t, then the resulting statement is a tautology.
- If $s \iff t$, you can always replace s with t in statement without changing the truth values.

Example 2.9. The statement $s : (p \to q) \leftrightarrow (\neg p \vee q)$ is a tautology. Replacing p with $(\neg r \wedge t)$ yields that

$$((\neg r \wedge t) \to q) \leftrightarrow (\neg(\neg r \wedge t) \vee q)$$

is also a tautology.

Example 2.10. By the law of double negation we have:

$$(p \to q) \to \neg p \iff (\neg\neg p \to q) \to \neg p.$$

Here is a more meaty example of the use of the Laws of Logic and the Rules of substitution.

Example 2.11. Show that $(\neg p \to q) \to r \iff (p \to r) \wedge (q \to r)$.

Solution

	$(\neg p \to q) \to r$	
$\iff$	$(\neg\neg p \vee q) \to r$	Substituting $\neg p$ for s and q for t into $s \to t \iff \neg s \vee t$
$\iff$	$(p \vee q) \to r$	Double Negation
$\iff$	$\neg(p \vee q) \vee r$	Subtitution into $s \to t \iff \neg s \vee t$
$\iff$	$(\neg p \wedge \neg q) \vee r$	DeMorgan
$\iff$	$r \vee (\neg p \wedge \neg q)$	Commutativity
$\iff$	$(r \vee \neg p) \wedge (r \vee \neg q)$	Distributivity
$\iff$	$(p \to r) \wedge (q \to r)$	$s \to t \iff \neg s \vee t$

There are a couple more statements that are related to the implication $p \to q$ which get special names as they are used often as proof techniques.

The *contrapositive* of $p \to q$ is $\neg q \to \neg p$. The *converse* of $p \to q$ is $q \to p$. The *inverse* of $p \to q$ is $\neg p \to \neg q$.

Problem 2.7. Show that an implication is logically equivalent to its contrapositive, but not to its converse or inverse. What does this tell you of the relation between the converse and the inverse of an implication?

In proving an implication, one common technique is to prove the contrapositive. When we prove an implication, we often comment on whether or not the converse is true.

Problem 2.8. Where p is the sentence 'It is raining', and q is the sentence 'I skip class', what are the English interpretations of $p \to q$, its contrapositive, its inverse, and its converse.

Let's finish this section with another example of proving the equivalence of statements using our laws of logic.

Example 2.12. Negate and simplify the statement $p \to (\neg q \wedge r)$.

Solution

Leaving the word 'simplify' a little vague, we proceed:

$$\begin{array}{lcl} & \neg(p \to (\neg q \wedge r) & \\ \iff & \neg(\neg p \vee (\neg q \wedge r) & \text{Using } s \to t \iff \neg s \vee t \\ \iff & \neg\neg p \wedge \neg(\neg q \wedge r) & \text{DeMorgan} \\ \iff & p \wedge (\neg\neg q \vee \neg r) & \text{Double Negation then DeMorgan} \\ \iff & p \wedge (r \to q) & \text{Using } s \to t \iff \neg s \vee t \end{array}$$

Observe that we used Double Negation and DeMorgan together in one step, and then in the next step, we used Double Negation without even mentioning it. We allow such shortcuts if it doesn't hinder reading. But don't take them too far.

The text finished this section with some examples of how a logical statement can be modelled by an electronic circuit. We won't look at it, but it you are so inclined, it is an interesting way to visualise the equivalence of statements.

2.3 Inference Rules

2.3.1 Logical Implication

Definition 2.13. An *argument* consists of a conjunction of statements :

$$p_1 \wedge p_2 \wedge \cdots \wedge p_d$$

called the *premises* and a *conclusion* q.

We write an argument as

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_d] \to q.$$

It is called true or *valid* if it is a tautalogy.

Example 2.14. Is the following argument valid?

$$[(p \to r) \wedge (\neg q \to p) \wedge \neg r] \to q$$

Solution

Again, we can use a truth table.

p	q	r	$s_1 : p \to r$	$s_2 : \neg q \to r$	$s_3 : \neg r$	$[s_1 \wedge s_2 \wedge s_3] \to q$
0	0	0	1	0	1	
0	0	1	1	1	0	
0	1	0	1	1	1	1
0	1	1	1	1	0	
1	0	0	0	0	1	
1	0	1	1	1	0	
1	1	0	0	1	1	
1	1	1	1	1	0	

There is only one row in which all of the premises are true, and in this case the conclusion is also true, so the argument is valid.

Example 2.15. Is the argument $[(p \to q) \wedge q] \to p$ valid?

Solution

No! (Don't you dare.) If $q = 1$ and $p = 0$ then $(p \to q) \wedge q$ is 1, but $p = 0$.

Example 2.16. Is the argument $[p \vee \neg p] \to q$ valid?

Indeed it is, as the premises are a contradiction. In this case we say that the argument is *vacuoulsy true*. 'Vacuous' means 'empty', like a vacuum. It is saying, 'sure the argument is true, but this is not really saying anything'.

Definition 2.17. If p and q are statements such that $p \to q$ is a tautology, then we say p *logically implies* q and write $p \Rightarrow q$

Note that $p \to q$ can be true or false, but $p \Rightarrow q$ means $p \to q$ is a tautology. So '$p \iff q$' is the same as '$p \Rightarrow q$ and $q \Rightarrow p$'.

Problem 2.9. Use a truth table to prove that $[(p \to q) \wedge p] \Rightarrow q$.

The logical implication in the above problem is a *inference rule* called *Modus Ponens*. Having proved it, we use it in proving the validity of other arguments.

Example 2.18. To prove the implication $[(p \to q) \wedge \neg q] \Rightarrow \neg p$ we start with the premises, and work to the conclusion.

$$\begin{array}{lcl} & (p \to q) \wedge \neg q & \\ \iff & (\neg q \to \neg p) \wedge \neg q & (contraposative) \\ \Rightarrow & \neg p & (ModusPonens). \end{array}$$

It will make an argument clearer if we write it in a table as follows.

1	$p \to q$	premise (given)
2	$\neg q$	premise
3	$\neg q \to \neg p$	contraposative of 1
$\therefore$	$\neg p$	2 and 3 and Modus Ponens

The symbol $\therefore$ is read 'therefore', and denotes that we have arrived at a conclusion. The implication above is another inference fule called *Modus Tollens*. We have several more inference rules:

	Premises	Conclusion	Name
1	$p \wedge (p \to q)$	q	Modus Ponens
2	$(p \to q) \wedge (q \to r)$	$p \to r$	Syllogism
3	$(p \to q) \wedge \neg q$	$\neg p$	Modus Tollens
4	$(p) \wedge (q)$	$p \wedge q$	Conjunction
5	$(p \vee q) \wedge \neg p$	q	Disjunctive Syllogism
6	$(\neg p \to F_0)$	p	Contradiction
7	$p \wedge q$	p	Conjunctive Simplification
8	p	$p \vee q$	Disjunctive Amplification
9	$(p \wedge q) \wedge [p \to (q \to r)]$	r	Conditional Proof
10	$(p \to r) \wedge (q \to r)$	$(p \vee q) \to r$	Proof by cases
11	$(p \to q) \wedge (r \to s) \wedge (p \vee r)$	$q \vee s$	Constructive Dilemma
12	$(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)$	$(\neg p \vee \neg r)$	Destructive Dilemma

These will help us prove logical implications.

Example 2.19. Show the argument $[(p \to q) \wedge \neg q \wedge \neg r] \to \neg(p \vee r)$ is valid.

Solution

1	$p \to q$	premise
2	$\neg q$	premise
4	$\neg p$	1 and 2 and Modus Tollens
5	$\neg r$	premise
6	$\neg p \wedge \neg r$	conjunction of 4 and 5
$\therefore$	$\neg(p \vee r)$	6 DeMorgan

Problem 2.10. Prove $[(p \to q) \wedge (r \to \neg q) \wedge r] \Rightarrow \neg p$.

Problem 2.11. Prove $[(p \to (q \to r)) \wedge (p \vee s) \wedge (t \to q) \wedge \neg s] \Rightarrow (\neg r \to \neg t)$.

To show that an argument is false, we need only to give a counterexample.

Example 2.20. To see that

1	$p \to q$
2	$\neg p$
$\therefore$	$\neg q$

is an invalid argument, let $p = 0$ and $q = 1$. Then premises 1 and 2 are true, but the conclusion is false.

2.3.2 Common Proof Techniques

Two common proof techniques that we get from our inference rules are 'Proof by Contradiction' and 'Indirect proof.' Proof by Contradiction us a proof that used the rule of contradiction. For this we introduce a false premise, the negation of our conclusion, and show it leads to falsity.

Example 2.21. We show that

1	$(s \wedge h) \to r$
2	$r \to c$
3	$\neg c$
$\therefore$	$\neg s \wedge \neg h$

is valid. Indeed, taking 1 - 3 as premises, we continues as follows

4	$\neg(\neg s \wedge \neg h)$	contradictory premise
5	$s \wedge h$	4 and DeMorgan
6	r	1 and 5 and MP (Modus Ponens)
7	c	2 and 6 and MP
8	$c \wedge \neg c$	3 conjunction 7
9	F_0	8 and Inverse Law
$\therefore$	$\neg s \wedge \neg h$	4 and 9 and contradiction

Since $(p \to (q \to r)) \iff ((p \wedge q) \to r)$, to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n] \to (q \to r)$$

it is enough to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge q] \to (r).$$

This is called an indirect proof.

Example 2.22. We prove the Rule of Syllogism (see the table) by indirect proof.

1	$p \to q$	premise
2	$q \to r$	premise
3	p	indirect premise
4	q	1,3, MP
$\therefore$	r	2,4 MP

Thus by indirect proof, we get the rule of Syllogism.

Problem 2.12. Prove

1	$p \to (q \to r)$
2	$\neg q \to \neg p$
$\therefore$	$p \to r$

2.4 Quantifiers

Recall that statements such as

$$p(x) : x \text{ is a multiple of } 3,$$

and

$$q(x, y) : x > y,$$

are not **statements**. Because of the variable, their truth values cannot be determined. However, when their variables are replaced with numbers, the become statements.

Definition 2.23. An *open statement* is a sentence that

- contains one or more variable,
- becomes a statement when the variables are replaced with certain allowable choices.

Example 2.24. Where $p(x)$ is the open statement above, $p(6)$ is the true statement '6 is a mutliple of 3', while $p(5)$ is the false statement '5 is a mutliple of 3'.

There is another way to change a open statement into a statement.

Definition 2.25. The allowable choices for the variables of an open statement are called the *universe* or *universe of discourse*, of the open statement. The symbol $\forall$, read as 'for every', is the *universal quantifier*. The symbol $\exists$, read as 'for some', is the *existential quantifiers*. They *quantify* an open statement $r(x)$ with universe U, as follows.

'$\forall x[r(x)]$' is the statement that '$r(x)$ is true for **all** $x \in U$.'

'$\exists x[r(x)]$' is the statement that '$r(x)$ is true for **some** $x \in U$.'

Note

Often, the universe of an open statement does not matter, or can be assumed from 'what is reasonable'. Certainly to interpret $p(6)$ as the statement '6 is a multiple of 3', we don't need to know the universe. But for quantified statements, it is usually important. Indeed, where $p(x)$ is as above $\forall x[p(x)]$ is false if the universe is $U = \mathbb{Z}$, but true if it is $U = 3\mathbb{Z}$.

That said, choosing the universe as $U = 3\mathbb{Z}$ seems a little like trickery. Sometimes even for open statements, if the universe is omitted, we assume it to be the largest reasonable universe. The statement 'x is a multiple of 3' appears very much to be about the integers, it would be reasonable to assume this is the universe.

Example 2.26. Let $q(x, y)$ be the statement $x > y$ have universe $\mathbb{R}$. Then

(a) $\forall x \forall y[q(x, y)]$ is false.

(b) $\forall x \exists y[q(x, y)]$ is true.

(c) $\exists x \forall y[q(x, y)]$ is false.

Hey look! The order of the quantifiers matters.

Problem 2.13. Where the universe is $\mathbb{Z}$, let

- $p(x) : x^2 - 5x + 6 = 0$
- $q(x) : x > 0$
- $r(x) : x^2 - 7 \geq 0$.

Determine the truth value of the following statements.

(a) $\exists x[p(x) \wedge \neg q(x)]$

(b) $\exists x[\neg q(x) \to p(x)]$

(c) $\forall x[p(x) \to q(x)]$

(d) $\forall x[p(x) \vee (q(x) \wedge r(x))]$

Problem 2.14. Where the universe of x is the set of students in the class, and the universe of y is a score in $s(x, y)$ be the open statement:

x got y on the test.

Hint

You can use arithmetic symbols $<$, $\leq$, and $=$.

Express the following statements symbolically.

(a) Some student got 7.5 on the test.

(b) Every student got at least 5 on the test.

(c) Some student got more than 9 on the test.

(d) No student got 8.5 on the test.

(e) Only one student got 0 on the test.

2.4.1 Quantifications of Logical Equivalences

Recall the logical equivalence

$$p \to q \iff \not{p} \vee q.$$

It follows by substitution that if $p(x)$ and $q(x)$ are open statements with the same universe, then

$$p(x) \to q(x) \iff \not{p}(x) \vee q(x)$$

for any choice of x in the universe.

Thus the following quantified versions of the logical equivalence are also true:

- $\forall x[p(x) \to q(x)] \iff \forall x[\neg p(x) \vee q(x)]$
- $\exists x[p(x) \to q(x)] \iff \exists x[\neg p(x) \vee q(x)]$

Every logical equivalence has similarily quantified versions.

2.4.2 Interaction of Quantifiers and Connectives

One can show the following.

Proposition 2.27. *(a)* $\forall x[p(x) \wedge q(x)] \iff \forall x[p(x)] \wedge \forall x[q(x)]$

(b) $\exists x[p(x) \wedge q(x)] \Rightarrow \exists x[p(x)] \wedge \exists x[q(x)]$

(c) $\forall x[p(x) \vee q(x)] \Leftarrow x[p(x)] \vee \forall x[q(x)]$

(d) $\exists x[p(x) \vee q(x)] \iff \exists x[p(x)] \vee \exists x[q(x)]$

(e) $\neg\forall x[p(x)] \iff \exists x[\neg p(x)]$

(f) $\neg\exists x[p(x)] \iff \forall x[\neg p(x)]$

Problem 2.15. Convince yourself that the logical implications from Proposition 2.27 are true. Show that $\Leftarrow$ is not true in (b) and $\Rightarrow$ is not true in (c).

The above rules allow us to prove other logical implications involving the quantifiers.

Example 2.28. Show that $\exists x[p(x) \to q(x)] \wedge \forall x[p(x)] \Rightarrow \exists x[q(x)]$.

Solution

$$\begin{array}{lll} & \exists x[p(x) \to q(x)] \wedge \forall x[p(x)] & \\ \iff & \exists x[\neg p(x) \vee q(x)] \wedge \forall x[p(x)] & \text{from } s \to t \iff \neg s \vee t \\ \iff & \exists x[\neg p(x)] \vee \exists x[q(x)] \wedge \forall x[p(x)] & \text{Prop 2.27 (d)} \end{array}$$